



PROJECT 7

EMPLOYEE SCORE VS. EMPLOYEE BACKGROUND

Linear Regression with Multiple Predictor

PRESENTED BY MUNAWAR

BACKGROUND

This analysis of Employee Performance data aims to delve deeper into the factors that influence individual evaluation scores (emp-scores) beyond just work hours. The primary focus of this study is to test the hypothesis regarding how educational level (emp-degree) is significantly related to the achievement of these performance scores (emp-scores).

EXPLORATORY DATA ANALYSIS

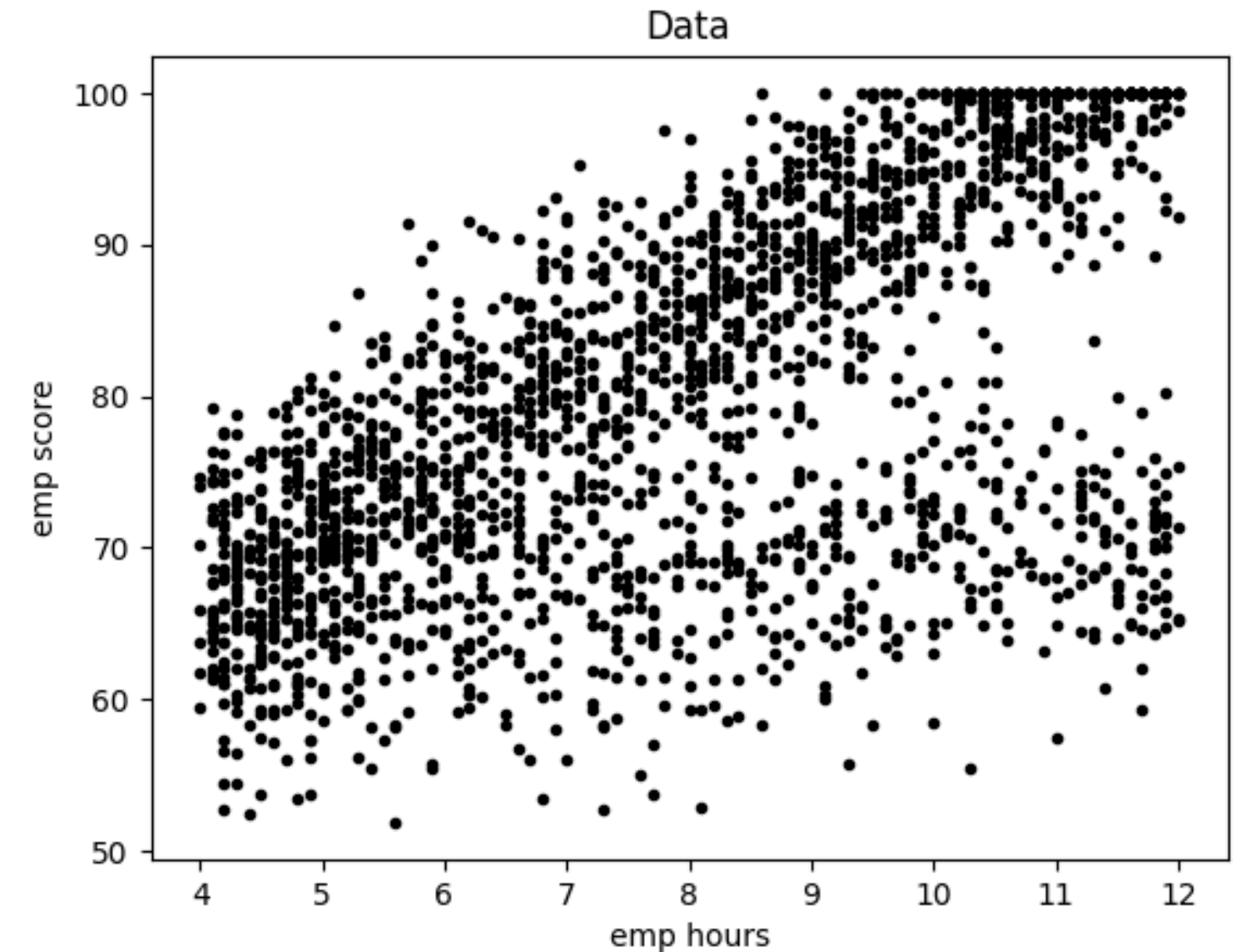
Statistik Descriptive

	emp_score	emp_degree	emp_experience	emp_hours	emp_age
count	2160.000000	2160.000000	2160.000000	2160.000000	2160.000000
mean	79.867454	0.683333	15.259259	7.995185	38.984722
std	12.541678	0.465284	8.805889	2.337024	12.452159
min	51.900000	0.000000	0.000000	4.000000	18.000000
25%	69.700000	0.000000	8.000000	5.900000	28.000000
50%	78.300000	1.000000	16.000000	8.100000	39.000000
75%	90.725000	1.000000	23.000000	10.000000	50.000000
max	100.000000	1.000000	30.000000	12.000000	60.000000

EXPLORATORY DATA ANALYSIS

Scatter Plot: emp_score vs emp_hours

- There is a positive pattern between working hours (emp_hours) and employee scores (emp_score).
- The data distribution shows two distinct groups based on emp_degree.
- Employees with degrees tend to have higher scores for the same number of hours worked.
- Strong indication of an interaction between emp_hours and emp_degree.



MODEL 1 - SIMPLE LINEAR REGRESSION USING CONTINUOUS VARIABLE

$$emp_{score} = 54.20 + 3.21emp_{hours}$$

Parameter	Estimate	Std Error	Interpretation
Intercept	54.20	0.7710	This value indicates that if an employee does not work at all or works zero hours (emp_hours = 0), then the estimated baseline performance score (emp_score) is 54.20. This can be considered a baseline score without the influence of working hours.
emp_hours	3.21	0.0925	This value indicates a positive relationship between working hours and performance scores. Specifically, for every additional hour worked, the employee's performance score is predicted to increase by 3.21 points.

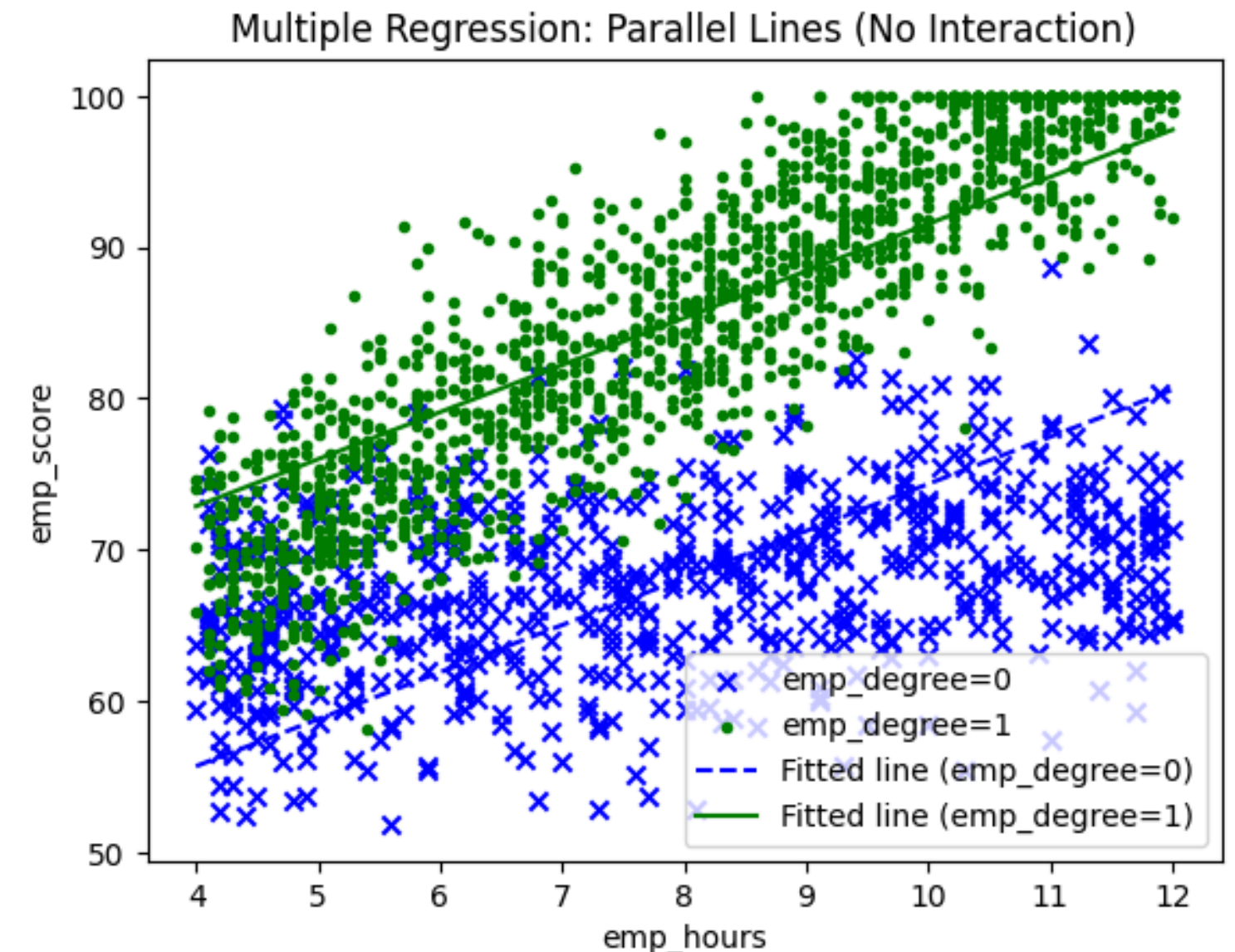
MODEL 2 - LINEAR REGRESSION USING MULTIPLE VARIABLE

$$\text{emp_score} = 43.270 + 3.11\text{emp_hours} + 17.17\text{emp_degree}$$

Parameter	Estimate	Std Error	Interpretation
Intercept	43.27	0.7710	The model predicts a score of 43.27 for an individual with a score of 0 for hours worked (emp_hours) and no degree (emp_degree)
emp_hours	3.11	0.0925	If we compare two individuals with the same degree status (emp_degree) but differing by 1 unit of work hours, the expected difference in scores is 3.11
emp_degree	17.17	0.2823	If we compare two individuals with the same number of hours worked (emp_hours) but differing in degree holdings, the expected difference in scores is 17.17

MODEL 2 - LINEAR REGRESSION USING MULTIPLE VARIABLE

- Based on this picture, the slope of the regression model between employee score (emp_score) and hours worked (emp_hours) is identical or parallel for both subgroups (with and without degrees).
- However, if we examine the data distribution in the graph, we see that the trend or slope of the data points actually shows a significant difference between the **emp_degree=0** and **emp_degree=1** groups.
- To address this, we need to include an interaction variable between **emp_degree** and **emp_hours**, which is the product of the two variables, as a new predictor.



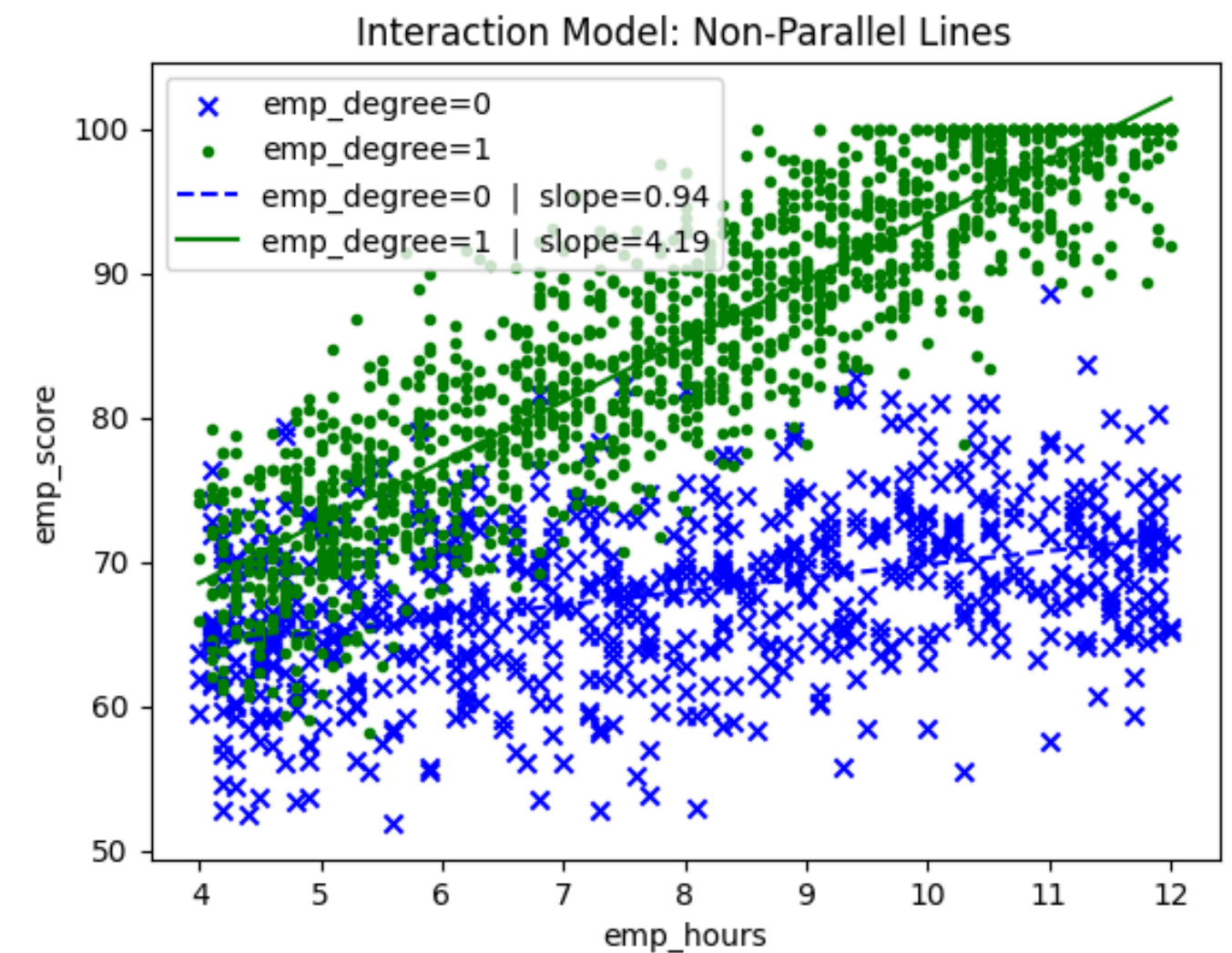
MODEL 3 - LINEAR REGRESSION USING MULTIPLE VARIABLE WITH INTERACTION

$$y = \beta_0 + \beta_1 \cdot emp_{hours} + \beta_2 \cdot emp_{degree} + \beta_3 \cdot (emp_{degree} \times emp_{hours})$$

By adding the interaction term **emp_degree:emp_hours**, the slope of **emp_hours** may **differ** between groups.

- $emp_degree = 0 \rightarrow \text{slope} = \beta_1$
- $emp_degree = 1 \rightarrow \text{slope} = \beta_1 + \beta_3$

if β_3 is significant $\rightarrow$ **there is an interaction** $\rightarrow$ **the two lines are not parallel**



MODEL 3 - LINEAR REGRESSION USING MULTIPLE VARIABLE WITH INTERACTION

$$emp_{score} = 60.42 + 0.94emp_{hours} - 8.70emp_{degree} + 3.26 (emp_{degree} \times emp_{hours})$$

Parameter	Estimate	Std Error	Interpretation
Intercept	60.42	0.6499	Represents the average employee score (emp_score) for those without a degree and working 0 hours—a scenario that may be practically meaningless without centering the predictors
emp_hours	0.96	0.0788	he comparison of average scores among employees without a degree who work 1 hour. This is the slope of the blue dotted line on the graph
emp_degree	-8.70	0.7997	The difference in predicted scores between employees with and without a degree when hours worked are 0. This coefficient is difficult to interpret directly due to its relationship to the interaction variable.
emp_hours:emp_degree	3.25	0.0964	Represents the difference in slopes for emp_hours when comparing employees with degrees to those without degrees

MODEL 3 - LINEAR REGRESSION USING MULTIPLE VARIABLE WITH INTERACTION

or simply can write

Has No Degree (emp_degree = 0)	Has Degree (emp_degree = 1)
emp_score = 60.42 + 0.94.emp_hours The estimated difference in scores of employees who do not have a degree for every 1 hour of work is 0.94.	emp_score = 51.72 + 4.20.emp_hours The estimated difference in scores of employees who have degrees for every 1 hour of work difference is 4.20.

Interaction coefficient (3.26) is significant → degree holders benefit MORE from each additional hour worked

MODEL COMPARISON

Model	Predictor	Intercept	emp_hours coef	emp_degree coef	Key Insight
Model 1 (Simple)	emp_hours	54.20	3.21	-	Hours → +3.21 per hr
Model 2 (Multiple)	emp_hours + emp_degree	43.27	3.11	+17.17	Degree add 17.17 pts
Model 3 (Interaction)	emp_hours + emp_degree + interaction	60.42	0.94 (no-deg) / 4.20 (deg)	-8.70(at 0 hr)	Slope differ

Key Takeaways

- Degree holder score 17.7 pts higher on average
- Interaction model reveals different productivity returns by education level

CONCLUSION

1

Education Matters

Employees with a degree consistently outperform non-degree holders by ~17 points on average.

2

Hours × Degree Interaction

Degree holders gain 4.20 pts per hour vs. 0.94 pts for non-degree; investing in skilled talent yields higher returns.

3

Model Selection

The interaction model is the most accurate. Assuming parallel slopes (Model 2) underestimates degree holders' productivity slope.

4

HR Recommendation

Encourage degree completion & reward high-hour commitment especially for educated staff to maximize performance scores.

**THANK YOU FOR
YOUR TIME**



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DATA ANALYST PRESENTATION
